

William Bekerman

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EDUCATION

University of Pennsylvania, The Wharton School, Philadelphia, PA

Expected December 2026

Doctor of Philosophy, Statistics

Honors: National Science Foundation Graduate Research Fellow

Committee: Dylan Small (advisor), Eric Tchetgen Tchetgen, Zhimei Ren

Cornell University, College of Agriculture and Life Sciences, Ithaca, NY

August 2018 – May 2022

Bachelor of Science, Biometry & Statistics (summa cum laude)

Honors: Merrill Presidential Scholar; Hunter R. Rawlings III Cornell Presidential Research Scholar

Advisor: Joe Guinness

PUBLICATIONS

Journal Articles

- **Bekerman, W.**, Dalal, A., del Ninno, C., & Small, D.S. (2026). Planning for gold: Hypothesis screening with split samples for valid powerful testing in matched observational studies. *Biometrika*, 113(1), p.asaf078.
- Mi, X., **Bekerman, W.**, Sims, P.A., Canoll, P.D., & Hu, J. (2024). RZiMM-scRNA: A regularized zero-inflated mixture model framework for single-cell RNA-seq data. *Annals of Applied Statistics*, 18(1), 1-22.
- **Bekerman, W.**, & Guinness, J. (2023). Comparison of CYGNSS and Jason-3 wind speed measurements via Gaussian processes. *Data Science in Science*, 2(1), 2194349.
- **Bekerman, W.**, & Srivastava, M. (2021). Determining decomposition levels for wavelet denoising using sparsity plot. *IEEE Access*, 9, 110582–110591.

Submitted/Revision

- **Bekerman, W.**, Mehta, A., Hasson, R.E., Robinson L.E., Small, D.S., & Fogarty C.B. (2026). Powerful Multivariate Sensitivity Analysis via Sample Splitting in an Observational Study of the Effects of Poverty on Cardiovascular Disease Risk Factors. *Submitted to Journal of the American Statistical Association*.
- Jin, H. A., **Bekerman, W.**, Small, D.S., & Rabinowitz, A. (2024). An observational study on effects of contact, collision, and non-contact sports participation on cognitive and emotional health. *Under Revision at PLOS One*.

Preprints

- Zhang, R., Kong, J., Small, D. S., & **Bekerman, W.** (2025). Beyond the ACE score: Replicable combinations of adverse childhood experiences that worsen depression risk. *arXiv:2511.19574*.
- Zhang, R., Kong, J., Small, D. S., & **Bekerman, W.** (2025). Protocol for an observational study on the effects of combinations of adverse childhood experiences on adult depression. *arXiv:2502.17679*.
- **Bekerman, W.**, Bogomolov, M., Heller, R., & Small, D.S. (2024). Protocol for an observational study on the effects of paternal alcohol use disorder on children's later life outcomes. *arXiv:2412.15535*.

PRESENTATIONS

- American Causal Inference Conference, Salt Lake City, UT. May 2026.
- European Causal Inference Meeting, Oxford, UK. April 2026.
- Online Causal Inference Seminar, Virtual Presentation. March 2026.
- International Conference on Multiple Comparison Procedures, Philadelphia, PA. August 2025.
- Joint Statistical Meetings, Nashville, TN. August 2025.
 - **IMS Hannan Graduate Student Travel Award Recipient.**
- American Causal Inference Conference, Detroit, MI. May 2025.
- ENAR Spring Meeting, New Orleans, LA. March 2025.

- Larry Brown PhD Student Symposium, Philadelphia, PA. November 2024
- International Seminar on Selective Inference, Virtual Presentation. October 2024.
- Penn Selective Inference Reading Group, Philadelphia, PA. October 2024, February 2025.
- Center for Causal Inference, University of Pennsylvania, Philadelphia, PA. September 2024, September 2025.
- Penn Conference on Big Data in Biomedical & Population Health Sciences, Philadelphia, PA. September 2024.
- American Causal Inference Conference, Seattle, WA. May 2024.
- Biophysical Society Annual Meeting, San Francisco, CA. February 2022.
- Joint Statistical Meetings, Virtual Conference. August 2021.

INDUSTRY EXPERIENCE & SELECTED PROJECTS

Pharmaceutical Consulting

2021

ZS Associates (Decision Analytics Intern)

- Optimized the efficiency of an omni-strategy messaging campaign for a large pharmaceutical company.
- Identified statistically significant engagement and retention discrepancies due to campaign overlap and messaging frequency.
- Developed and presented detailed, action-oriented recommendations during various client meetings.

NFL Big Data Bowl

2020

Cornell Data Science (Project Lead)

- Predicted the result of rushing plays using game, play, and player data provided by NFL's Next Gen Stats.
- Constructed a convolutional neural network model using the Keras library in Python and achieved a mean absolute error of approximately 2.35 yards and a cumulative rank probability score (CRPS) of approximately 0.0118 (top 1% in competition).
- Implemented a gradient boosting machine and random forest regression model, improving mean absolute error and CRPS by almost 15% compared to a baseline linear regression model.
- Visualized the data provided by NFL's Next Gen Stats and the outcome of each play using JavaScript.

Fake News Challenge

2019

Cornell Data Science

- Built a random forest model based on bag of words features to determine the relevance of an article for the Fake News Challenge.
- Constructed a siamese neural network model to simultaneously analyze the headline and body of an article to discern its stance.
- Achieved a relevance detection accuracy score of over 95% and stance detection accuracy of almost 75% (top ten post hoc).
- Created interactive visualizations of relevance/stance detection features and LSTM activations overlaid onto source text.
- *** Awarded First Prize (\$750) by Sandia National Laboratories at Cornell University's Bits on Our Mind Showcase.

UNDERGRADUATE TEACHING EXPERIENCE

- BTRY 4030 - Linear Models with Matrices. Fall 2021
- BTRY 6020 - Statistical Methods II (Graduate-Level). Spring 2021
- INFO 2950 - Introduction to Data Science. Fall 2020
- PLBIO 2400- Green World, Blue Planet. Fall 2019, Fall 2020, Fall 2021

SERVICE AND LEADERSHIP

Internal: Wharton Statistics Student Seminar Co-Organizer

External: Inaugural Student Representative, Executive Board of Society for Causal Inference (SCI)

Mentoring:

- PhD Students: Zihan Zhu (Penn)
- Undergraduate Students: Hannah Jin (Penn, now PhD student at Harvard), Matthew Spivey (Penn), Ruizhe Zhang (Fudan, now PhD student at Stanford)
- Served as peer advisor of students in Biometry & Statistics Major at Cornell

Reviewing: IEEE Transactions on Information Theory, Journal of Biopharmaceutical Statistics, Journal of Quantitative Analysis in Sports, Journal of the Royal Statistical Society (Series B)

Judging: Long Island Science and Engineering Fair: Intelligent Machines, Mathematics, Software, Robotics

Other:

- Alpha Phi Omega Service Fraternity (Family Head, Leadership/Friendship/Service Award Recipient)
- Cornell Data Science Project Team (Team Lead, Project Lead, Recruitment Chair)